

INTELLIQUERY: AI-POWERED CONVERSATIONAL BUSINESS INTELLIGENCE DASHBOARD PLATFORM

Proposal

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List of Abbreviations

AI	Artificial Intelligence
BI	Business Intelligence
NLP	Natural Language Processing
LLM	Large Language Model
DAX	Data Analysis Expressions
API	Application Programming Interface
SQL	Structured Query Language
UI/UX	User Interface/User Experience
REST	Representational State Transfer
BERT	Bidirectional Encoder Representations from Transformers
GPT	Generative Pre-trained Transformer
LSTM	Long Short-Term Memory

Abstract

IntelliQuery is an AI-powered conversational dashboard platform that revolutionizes business intelligence accessibility through natural language interaction. The system integrates with Power BI data sources but presents insights through a custom-built intelligent dashboard where users can ask questions in plain English and receive dynamic visualizations, reports, and actionable insights. Using advanced NLP, contextual memory management, and machine learning algorithms, the platform eliminates technical barriers to data analysis while providing enterprise-grade security and scalability. The project addresses the critical problem of democratizing business intelligence access for non-technical users in Pakistani enterprises, with potential applications across multiple industries. Built in collaboration with Folio3 Software Solutions, the platform combines cutting-edge agentic AI architecture with practical business intelligence needs.

Keywords: Conversational AI, Business Intelligence, Custom Dashboard, Natural Language Processing, Data Visualization, Enterprise Analytics, Machine Learning, Context Management, Agentic AI, LangGraph

1 Introduction and Problem Statement

1.1 The Problem We're Solving

Business intelligence tools are essential for modern organizations, but they create significant accessibility barriers that prevent effective data-driven decision making. Research indicates that 85% of business users struggle with traditional BI tools requiring SQL/DAX knowledge [1], while analysts spend 60–80% of their time on data preparation and basic query creation rather than analysis [2].

1.1.1 Primary Problems

1. **Technical Complexity:** Non-technical stakeholders depend entirely on IT teams for data insights
2. **Time Inefficiency:** Manual report generation creates bottlenecks and delays
3. **Limited Accessibility:** Static dashboards cannot answer dynamic business questions
4. **Context Loss:** Users lose analytical thread when navigating between different tools

1.1.2 Pakistani Enterprise Context

Pakistani enterprises face additional challenges including limited BI expertise, high costs of specialized consultants, and need for culturally adapted solutions that understand local business terminology and practices.

1.2 Why a Digital Solution is Essential

Traditional approaches cannot solve these problems effectively. Training all employees on complex BI tools is prohibitively expensive, and hiring additional analysts does not scale well with the growing complexity of organizational data. Manual report generation often creates bottlenecks that delay decision-making, while generic AI assistants lack the ability to understand domain-specific business context. These limitations make it clear that a specialized digital solution is necessary to ensure scalability, efficiency, and meaningful insights for enterprise users.

Our digital solution provides scalability, consistency, 24/7 availability, learning capability, and cost efficiency that manual processes cannot achieve.

1.3 Project Objectives

The primary objective of this project is to design and implement a conversational business intelligence platform that enables non-technical users to access and analyze organizational data seamlessly. Specifically, the system aims to:

1. **Develop an intelligent conversational interface** that understands domain-specific business terminology and context, ensuring that queries can be expressed in natural language without requiring technical expertise.
2. **Build a custom analytics dashboard** optimized for conversational interaction, where insights are presented in an interactive and visually intuitive manner, rather than relying on static BI dashboards.
3. **Implement an advanced NLP pipeline** capable of interpreting complex business queries, resolving ambiguities, and transforming user requests into accurate analytical outputs.
4. **Integrate seamlessly with existing Power BI infrastructure**, leveraging enterprise data models and security protocols while maintaining compatibility with organizational workflows.
5. **Enable dynamic visualization generation** so that users receive the most appropriate charts, tables, or metrics tailored to the nature of their queries and data characteristics.
6. **Create a contextual memory system** that preserves conversation history and analytical continuity, allowing users to refine queries, ask follow-ups, and explore data progressively without losing context.

1.4 Constraints

The development of IntelliQuery faces several practical and technical constraints that must be carefully managed:

- **Power BI API Limitations:** The platform is subject to API rate limits and complex authentication flows, which may restrict real-time data access and require optimized query management.
- **Real-Time Performance Requirements:** Since users expect instant responses, the system must achieve sub-second query processing and visualization rendering, even under high traffic.

- **Data Privacy and Security Compliance:** The system must adhere to strict enterprise data protection standards, ensuring encrypted storage, secure API communication, and compliance with regulations such as GDPR and local laws.
- **Academic Timeline and Resources:** The project must be completed within the academic cycle with limited team resources, requiring strict prioritization of features and efficient project management.

2 Literature Review

2.1 Natural Language to Database Query Systems

Recent research in text-to-SQL has shown significant progress with transformer-based models. The Spider dataset introduced by Yu et al. (2018) established an important benchmark for cross-domain semantic parsing, achieving strong accuracy on complex queries [3]. Large language models such as GPT-3 and GPT-4 have demonstrated few-shot learning capabilities for generating SQL queries [4], significantly reducing the need for handcrafted rules. Beyond academic datasets, enterprise-focused studies have highlighted the additional challenges of query disambiguation, schema alignment, and domain adaptation in business intelligence environments [5].

2.2 Conversational AI for Enterprise Applications

The integration of conversational AI into enterprise systems has been an active research area. Zhang et al. (2022) proposed advanced context management techniques for multi-turn dialogue, enabling improved handling of long conversation threads in enterprise scenarios [6]. Similarly, Liu et al. (2021) established design best practices for enterprise conversational systems, emphasizing usability, trust, and domain adaptation. Memory-augmented neural networks and vector-based semantic retrieval methods have further enhanced the ability to maintain context over extended analytical workflows, which are particularly relevant to business intelligence systems [7].

2.3 Research Gaps

Despite these advancements, most existing research primarily focuses on generic SQL generation tasks rather than business intelligence contexts, where queries involve complex relationships, hierarchical data, and nuanced business logic. Limited academic work specifically addresses conversational interfaces for democratizing BI in developing economies, where cultural, linguistic, and infrastructural factors create additional adoption challenges [5]. This gap highlights the need for domain-specific conversational BI systems that combine natural language processing with enterprise-grade security and usability.

3 Requirements and Features

3.1 Core AI Features

The system incorporates several advanced AI-driven capabilities that enable seamless, intelligent interaction with business data:

1. **Natural Language Query Processing** Translates user queries expressed in plain English into structured Power BI commands. Key aspects include:
 - Conversion of business questions into structured queries
 - Support for complex analytical requests (comparisons, trends, drill-downs)
 - Clarification dialogue for ambiguous queries
 - Confidence scoring to indicate interpretation reliability
2. **Dynamic Visualization Generation** Automatically generates appropriate and interactive visualizations in response to user queries. Capabilities include:
 - Intelligent chart selection based on data characteristics
 - Interactive features such as cross-filtering
 - Support for multiple visualization formats (bar, line, pie, tabular)
 - Real-time dashboard updates and modifications
3. **Contextual Memory Management** Maintains continuity and relevance in multi-turn conversations by:
 - Preserving conversation history across sessions
 - Resolving references in follow-up questions
 - Offering context-aware query suggestions
 - Managing memory persistence with pruning algorithms

3.2 User Interface Features

The user interface is designed for simplicity, responsiveness, and accessibility across platforms:

1. Modern, responsive web-based chat interface
2. Dual response modes: text explanations and visual insights

3. Real-time typing indicators and response streaming
4. Mobile-responsive design for cross-device access
5. Seamless dashboard integration with export options (PDF, Excel, PowerPoint)

3.3 User Flow Diagram

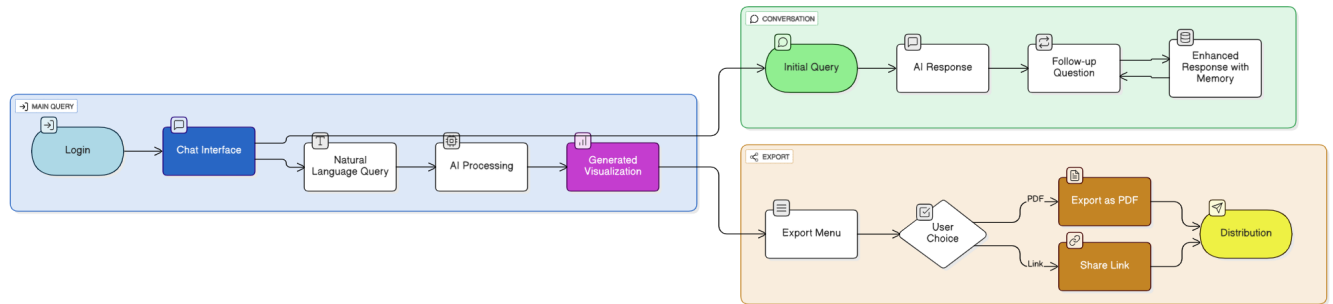


Figure 3.1: Comprehensive user flow from login to insight generation

The user flow illustrates three primary paths:

1. **Business User Query Flow:** Login → Chat Interface → Natural Language Query → AI Processing → Visualization → Follow-up
2. **Context-Aware Conversation:** Initial Query → Response → Follow-up Question → Enhanced Response → Memory Update
3. **Export and Sharing:** Generated Insight → Export Options → Distribution to Team

4 Technical Architecture

4.1 System Architecture Overview

The platform follows a microservices architecture with four primary layers:

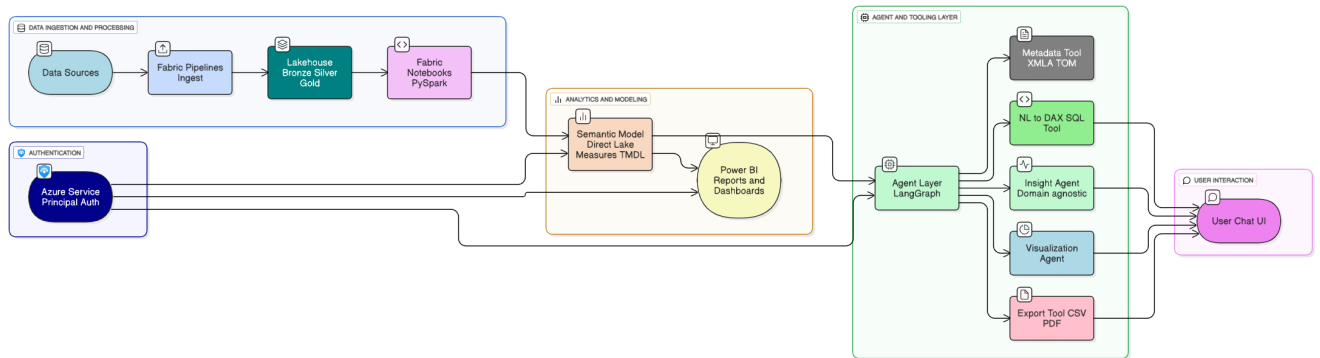


Figure 4.1: Comprehensive system architecture showing all components and data flow

1. **Frontend Layer:** React/Next.js with real-time WebSocket connections
2. **API Gateway Layer:** FastAPI with authentication and rate limiting
3. **Business Logic Layer:** Microservices for NLP, context management, and visualization
4. **Data Layer:** Multiple databases and Power BI integration

4.2 Custom Dashboard Architecture

The dashboard architecture is designed to provide an intuitive and efficient user experience, optimized for AI-driven interactions:

- **Conversation-Optimized Layout:** A user interface tailored for seamless AI interaction and intuitive navigation.
- **Dynamic Content Generation:** Charts and visualizations are created and updated in real-time, with adaptive layout adjustments.
- **Context-Aware Interface:** The UI intelligently responds to the flow of the conversation, prioritizing relevant data and insights.
- **Multi-Modal Export:** Supports exporting dashboards in multiple formats, including PDF, Excel, and PowerPoint.

5 Team Roles

Primary Responsibilities	Technical Components	Team Member
Frontend Development & UI/UX Design	• React/Next.js dashboard development • Chat interface implementation • Visualization components (Plotly.js) • Responsive design and mobile optimization • User experience testing	Izma Sameed
Power BI Integration	• API Bridge • Dataset Queries	Sameed Umair
AI/ML Development & NLP Pipeline	• Natural language processing implementation • Machine learning model development • Context management algorithms • LangGraph orchestration • Model training and evaluation	Izma Sameed Umair
Backend Development & Architecture	• FastAPI development and microservices • Database design (PostgreSQL/Redis) • Power BI API integration • Authentication and security systems • Performance optimization and caching	Izma Umair
Documentation & Demo	• Setup guide • User manual • Video	Izma Sameed Umair

Table 5.1: Detailed team member responsibilities

6 System Design

6.1 User Interface Mockup



Figure 6.1: Main dashboard interface showing chat integration with dynamic visualizations

The dashboard mockup shows the proposed UX: conversational chat on the left and dynamic visualizations on the right. It demonstrates how natural language interactions update the dashboard, present explanations, and offer export/share options for business users.

6.2 UML Diagrams

6.2.1 Use Case Diagram

The use case diagram highlights the overall functionality of IntelliQuery. It identifies the actors and the actions they can perform, such as submitting queries, receiving visualizations and managing conversations.

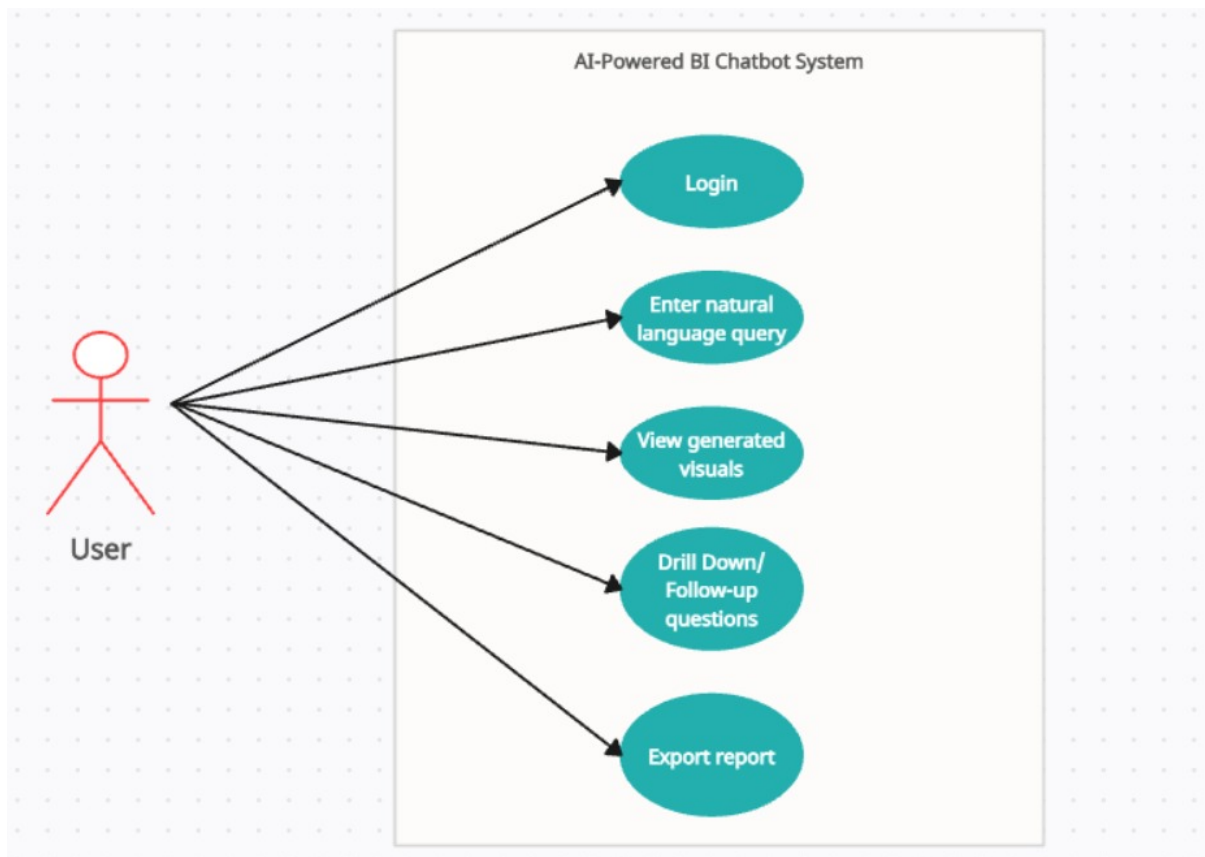


Figure 6.2: Use case diagram showing the primary interactions between users and the system

6.2.2 Activity Diagram

The activity diagram demonstrates the end-to-end user-focused workflow: the user logs in, submits natural language queries, the system validates or requests clarification, and successful queries produce visualizations that may be exported or shared. Decision points (e.g., query validation) are emphasized to show business logic flow.

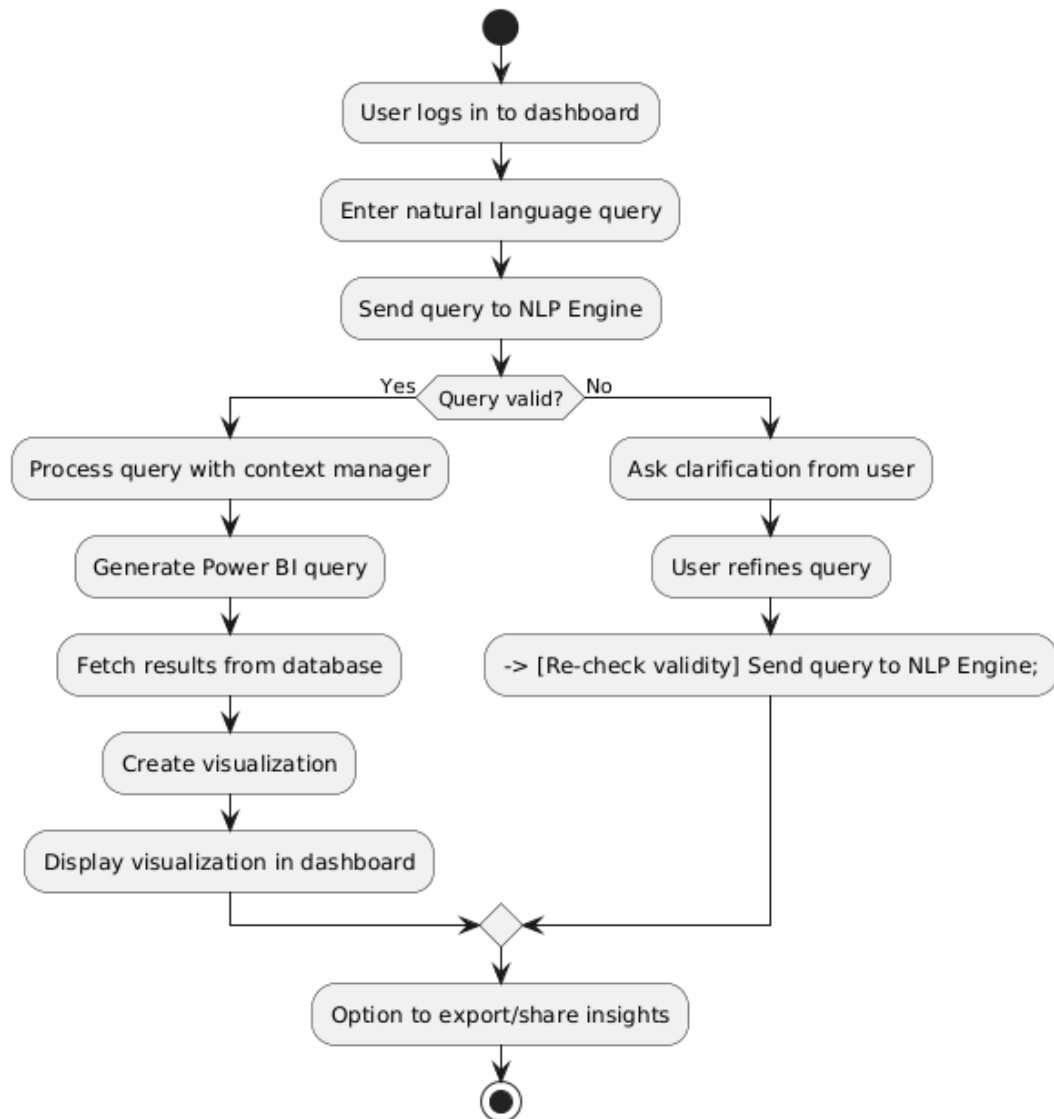


Figure 6.3: Activity diagram illustrating the end-to-end workflow of a user query

6.2.3 Sequence Diagram

The sequence diagram illustrates how different system modules communicate to process a query. It focuses on internal message exchanges between the frontend, NLP engine, context manager, Power BI API, and visualization renderer, highlighting system integration and response flow.

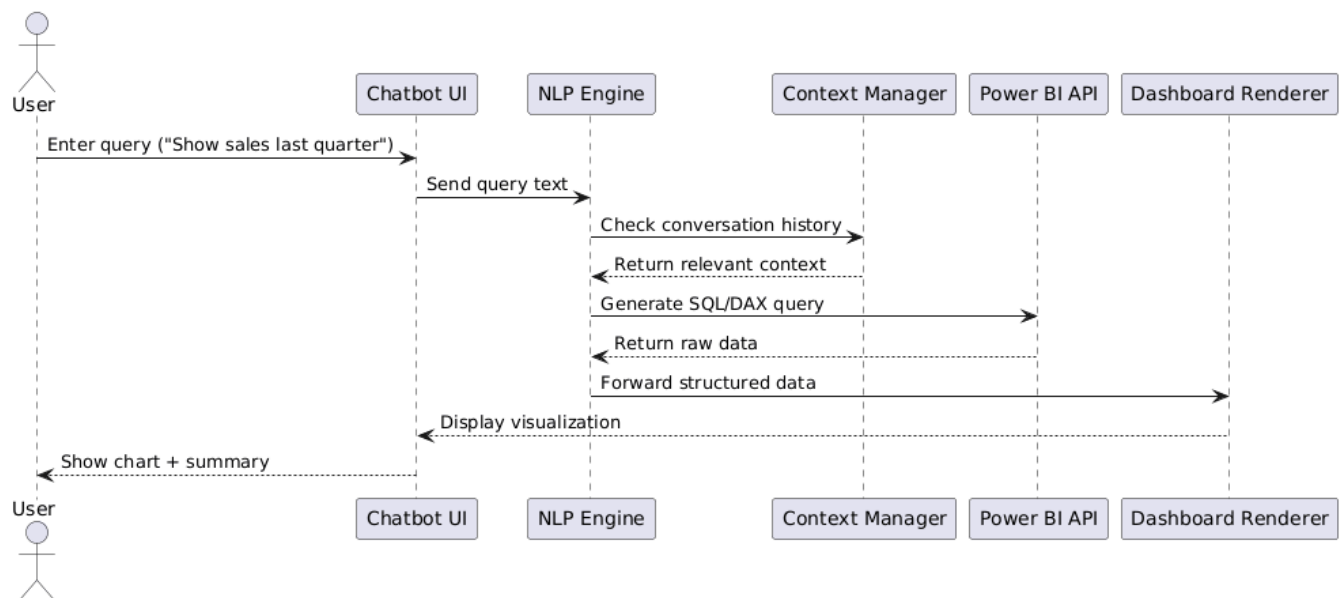


Figure 6.4: Sequence diagram showing technical interactions between system components

6.2.4 Deployment Diagram

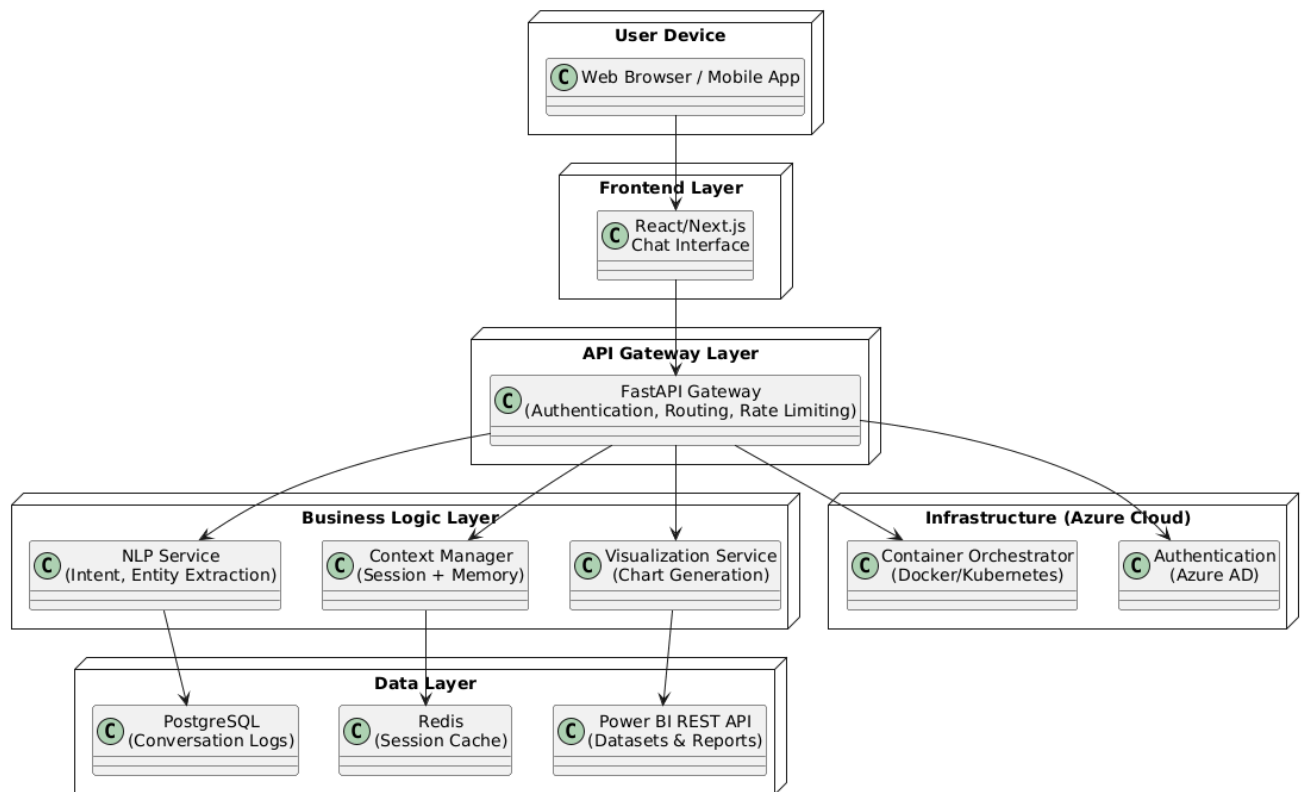


Figure 6.5: Deployment diagram showing physical distribution of system components

The deployment diagram shows how IntelliQuery’s components are physically deployed. It illustrates the interaction between the user interface (browser/mobile), the backend microservices (FastAPI, NLP engine, context manager), the data sources (Power BI, PostgreSQL, Redis), and the cloud infrastructure (Azure). This provides a clear picture of how the system will run in a real-world enterprise setting.

7 Data Structures

In the AI-powered conversational BI system, several data structures are critical to enabling the agentic workflow, context-aware query processing, and dynamic dashboard updates. Among these, three structures are essential and heavily used within this project: graphs, queues, and hash maps.

7.1 Graph Structures

Graphs are extensively used to model relationships between user queries, extracted entities, and visualization components. In this project:

- **Query-Entity Graph:** Nodes represent entities recognized in queries, while edges represent semantic relationships between them. This allows the NLP and API agents to resolve complex queries involving multiple related entities.
- **Visualization Dependency Graph:** Nodes correspond to charts or dashboard components, and edges encode dependencies based on data sources. The Visualization Agent uses this to determine which charts need updates when a query is executed.
- Graph traversal algorithms are applied to propagate context and maintain coherence across sequential queries, ensuring prior information is available for context-aware responses.

7.2 Queue Structures

Queues are used to manage the sequential flow of queries across agents:

- The **Query Queue** stores incoming user queries in a FIFO manner, guaranteeing that queries are processed in the order they are received.
- Each agent (NLP, Validation, Memory, API) dequeues tasks from this structure, ensuring coordinated processing and maintaining consistency in the workflow.
- Queues also allow buffering during high loads, preventing bottlenecks and ensuring the dashboard remains responsive even with multiple concurrent users.

7.3 Hash Maps / Dictionaries

Hash maps provide efficient key-value storage and retrieval, crucial for real-time processing in the project:

- The **Entity-Value Map** stores entities extracted from queries along with their resolved values, allowing the Validation and API agents to access parameters quickly.
- The **Agent State Map** keeps track of each agent's current context, processed tasks, and intermediate results, enabling smooth coordination between agents.
- Hash maps are also used in caching visualization results, minimizing repeated computations and improving dashboard update speed.

7.4 Summing Them Together

In this project, graphs, queues, and hash maps form the computational backbone, coordinating the agentic workflow:

1. User queries enter the **Query Queue** to preserve processing order.
2. NLP Agent dequeues the query, extracts intents and entities, and adds them to the **Query-Entity Graph**.
3. Validation Agent checks and updates entity values using **hash maps**.
4. Memory Agent retrieves context from the graph and stores intermediate states in hash maps.
5. API Agent fetches results, while the **Visualization Dependency Graph** determines which dashboard panels to update.
6. Each new query repeats this cycle, ensuring context-aware, responsive, and efficient dashboard updates.

8 Implementation Methodology

8.1 Development Phases

Phase	Objectives	Duration
Foundation	Basic chat interface, Power BI connection, core database setup	Weeks 1-6
Intelligence	NLP pipeline, context management, intent classification	Weeks 7-12
Advanced Features	Dynamic visualizations, memory management, performance optimization	Weeks 13-18
Testing & Deployment	System testing, user acceptance, documentation	Weeks 19-22

Table 8.1: Development phase breakdown

8.2 Technology Stack

- **Frontend:** React 18, Next.js 13, TypeScript, Plotly.js, Tailwind CSS
- **Backend:** Python 3.11, FastAPI, PostgreSQL, Redis, Docker
- **AI/ML:** OpenAI GPT-4, Custom BERT models, LangGraph, spaCy
- **Infrastructure:** Azure Cloud Services, Power BI REST API, WebSocket
- **Testing:** Jest, Pytest, Selenium, Load testing tools

8.3 Agentic Workflow

The agentic workflow describes the sequential interaction between user queries and the AI-driven dashboard system:

1. The user submits a query through the chat interface.
2. The NLP Agent interprets the input and extracts the user's intent.
3. The Validation Agent ensures schema alignment and corrects any input errors.
4. The Memory Agent incorporates context from prior queries to maintain conversational continuity.
5. The API Agent executes the query via the Power BI REST API.
6. The dashboard dynamically updates to reflect the results in real time.

8.4 Algorithmic Development

The algorithmic development outlines the core computational logic enabling the AI-powered dashboard:

1. Semantic parsing of natural language queries to extract structured meaning.
2. Agent-based intent classification and entity extraction for precise query understanding.
3. Queue-based contextual memory management to maintain continuity across interactions.
4. Rule-based error correction, with fallback to large language models for robust handling.
5. Visualization recommendation logic to suggest optimal charts and layouts based on query context.

8.5 Quality Assurance Strategy

1. **Unit Testing:** Individual component validation with 90%+ coverage
2. **Integration Testing:** End-to-end workflow validation
3. **Performance Testing:** Load testing with 50+ concurrent users
4. **Security Testing:** Data protection and authentication validation
5. **User Acceptance Testing:** Business scenario validation with Folio3

9 Evaluation and Testing

9.1 Success Metrics

Category	Metric	Target
Technical Performance	Query understanding accuracy	> 90%
	Response time (simple queries)	< 3 seconds
	Response time (complex visualizations)	< 10 seconds
	System availability	> 99%
	Concurrent user support	50+ users
User Experience	Task completion rate	> 85%
	User satisfaction score	> 4.2/5.0
	Learning curve	< 15 minutes
	Error recovery rate	> 90%
Business Impact	Time savings vs manual analysis	60-70%
	User adoption rate	> 70%
	Insight generation frequency	3x increase

Table 9.1: Comprehensive success metrics and targets

9.2 Evaluation Methodology

1. **A/B Testing:** Compare conversational vs traditional BI interfaces
2. **User Studies:** Task-based evaluation with business users from Folio3
3. **Performance Benchmarking:** Load testing under various conditions
4. **Accuracy Assessment:** Manual validation of generated insights against expected results

10 Timeline and Milestones

10.1 Gantt Chart

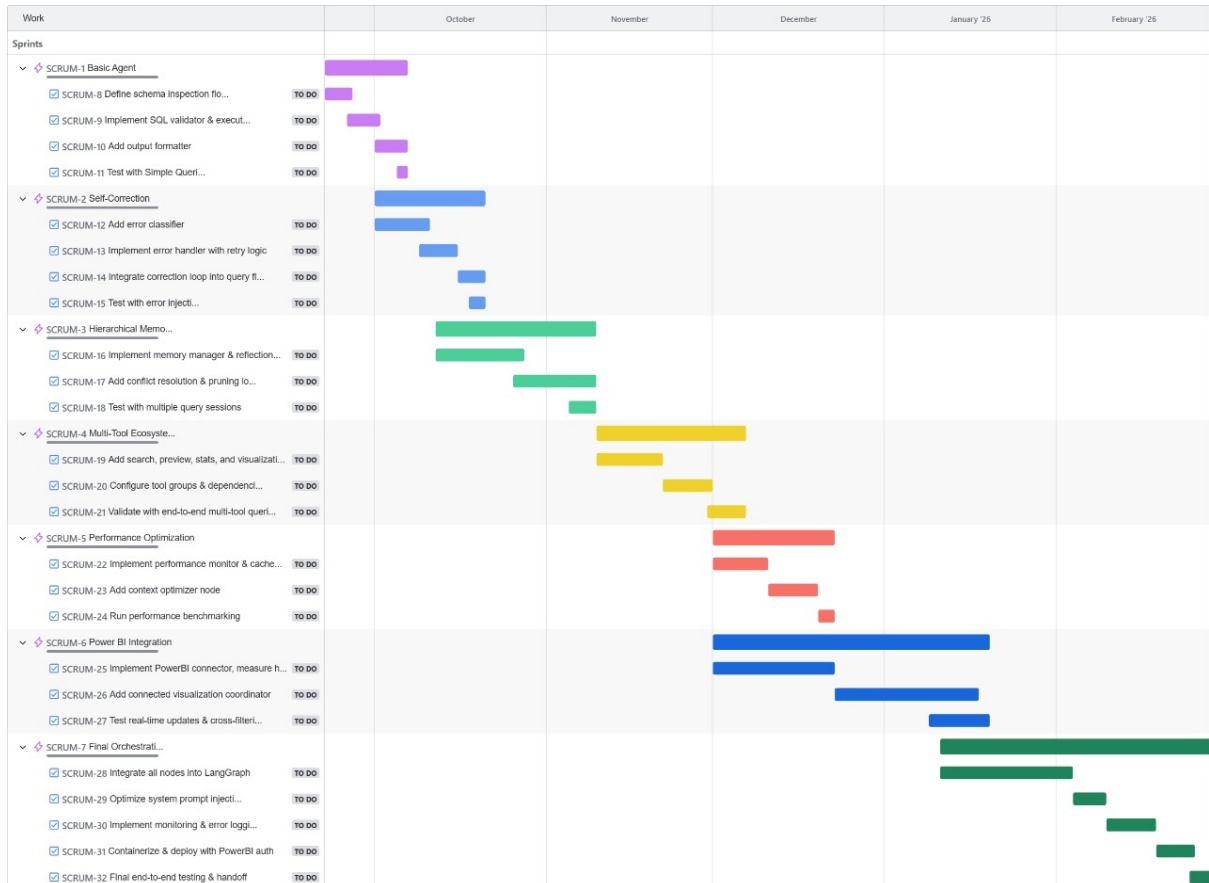


Figure 10.1: Detailed project timeline with dependencies and milestones on jira

10.2 Critical Milestones

1. **Week 6:** Basic conversational interface with Power BI connection demonstrated
2. **Week 12:** Full AI pipeline with dynamic visualization capabilities
3. **Week 18:** Complete system with performance optimization and advanced features
4. **Week 22:** Production-ready system with comprehensive documentation and testing

11 Market Analysis

11.1 Current Market Offerings

The conversational business intelligence (BI) market features several tools with varying capabilities. Power BI Copilot and Tableau Ask Data provide foundational AI functionality, but their features are limited, and they offer no or very limited dashboard customization. ThoughtSpot delivers a similar level of AI with no dashboard customization and relies on third-party integration for Power BI. All three are positioned as premium or high-cost solutions.

11.2 Our Differentiation: IntelliQuery

IntelliQuery stands out by offering a unique combination of advanced features that directly address the shortcomings of competitors. It provides advanced contextual memory and the ability to create fully customizable dashboards, empowering users with intuitive and efficient data interaction.

Unlike its rivals, IntelliQuery is being developed with planned voice integration and offers deep native Power BI integration, ensuring a seamless user experience. All of this is provided at an enterprise-friendly price, offering a superior value proposition compared to the costly, less flexible alternatives on the market.

11.3 Comparison Table

Feature	IntelliQuery	Power BI Copilot	ThoughtSpot	Tableau Ask Data
Contextual Memory	Advanced	Basic	Limited	Basic
Custom Dashboard	Full capability	Limited	No	No
Voice Integration	Planned	No	No	No
Cost-Effectiveness	Enterprise-friendly	Premium	High cost	High cost
Power BI Integration	Deep native	Native	Third-party	Third-party

Table 11.1: Feature comparison between IntelliQuery and competing solutions

11.4 Market Opportunity

The conversational business intelligence (BI) market is experiencing rapid growth, with a projected compound annual growth rate (CAGR) of 15

- **The Rise of Self-Service Analytics:** A growing number of enterprises are shifting toward self-service models, empowering employees at all levels to access and analyze data without relying on specialized teams.
- **Widespread AI Adoption:** The integration of conversational AI is becoming a standard feature in enterprise applications, making it a natural and expected component of modern data analytics tools.
- **An Expanding User Base:** With over 200 million users globally, Microsoft Power BI's massive and continuously growing ecosystem presents a significant, untapped market for powerful conversational solutions.
- **The Need for Data Democratization:** There is a pressing need to make data more accessible to users worldwide, particularly in developing markets where access to specialized analytics expertise may be limited.

12 Ethical Considerations

12.1 Data Privacy and Security

- **Robust Encryption:** Implement end-to-end encryption to protect sensitive business data in transit and at rest.
- **Regulatory Compliance:** Adhere to global data protection standards, including GDPR and relevant local privacy laws.
- **Secure Authentication:** Utilize enterprise-grade security protocols like Azure Active Directory for secure user authentication.
- **Proactive Auditing:** Conduct regular security audits and penetration testing to identify and address vulnerabilities.

12.2 Bias and Fairness

- **Bias Detection:** Continuously test for and mitigate bias in query interpretation and system responses.
- **Inclusive Training Data:** Use diverse training datasets to prevent sector-specific and other inherent biases.
- **Discriminatory Pattern Monitoring:** Actively monitor system outputs for any discriminatory patterns or unfair results.
- **Feedback Mechanisms:** Provide clear channels for users to report potential bias, facilitating rapid correction.

13 Risk Analysis

13.1 Technical Risks and Mitigation

- **LLM Integration Complexity:** Mitigated through extensive, modular testing and the implementation of robust fallback mechanisms.
- **Power BI API Limitations:** Addressed by intelligent data caching and comprehensive error handling to ensure reliability.
- **Performance Scalability:** Managed by designing a microservices architecture and implementing load balancing for high performance.
- **Data Security Concerns:** Mitigated through end-to-end encryption and strict, role-based access controls.

13.2 Project Management Risks

- **Scope Creep:** Prevented through a clear Minimum Viable Product (MVP) definition and regular stakeholder reviews.
- **Timeline Constraints:** Managed with an agile development approach and continuous feature prioritization.
- **Team Coordination:** Ensured through regular meetings, structured communication, and the use of collaborative development tools.

14 Deliverables

14.1 Primary Deliverables

1. A complete conversational dashboard platform with fully integrated AI capabilities.
2. A custom web-based interface optimized for intuitive business intelligence interaction.
3. A robust Power BI integration layer with secure, authorized data access.
4. Comprehensive project documentation, including technical specifications and user guides.
5. A professional demonstration video showcasing the system's full range of capabilities.

14.2 Technical Artifacts

1. The full source code with detailed documentation and a complete suite of tests.
2. Database schemas, deployment scripts, and configuration files.
3. Trained machine learning models and the datasets used for their development.
4. Performance benchmarks and a detailed analysis of evaluation results.
5. A full security assessment and documentation of compliance measures.

15 Industry Partnership

15.1 Folio3 Collaboration Benefits

Our collaboration with Folio3 Software Solutions provides a significant strategic advantage. It grants us access to critical enterprise BI requirements and real-world use cases, offering a valuable source of industry expertise. Furthermore, we receive professional guidance on system architecture and development best practices through their technical mentorship. The partnership also ensures the implementation of rigorous enterprise-grade development and testing methodologies, enhancing our quality assurance. Finally, their insights into commercial viability and evolving customer needs provide essential market validation for our project.

15.2 Professional Development

This collaboration offers invaluable opportunities for professional development. We will gain hands-on experience by working directly with seasoned industry professionals, while also receiving exposure to a professional enterprise development environment. This unique setting provides essential networking opportunities that will be vital for future career prospects. Ultimately, this experience will allow us to develop a strong portfolio of work to showcase our skills and expertise to potential employers.

16 Research Paper Potential

The proposed AI-powered conversational BI system demonstrates significant potential for research contributions. Key aspects that could be developed into a research paper include:

- **Agentic Architecture:** The use of multiple specialized agents (NLP, Validation, Memory, API) in a coordinated workflow offers a novel approach to conversational analytics.
- **Context-Aware Dashboarding:** Dynamic integration of context from previous queries to inform dashboard updates can be explored for performance evaluation and user experience improvements.
- **Algorithmic Innovations:** Semantic parsing, agent-based intent extraction, queue-based memory management, and visualization recommendation logic present opportunities for methodological contributions.
- **Comparative Evaluation:** The system can be benchmarked against existing conversational BI tools to quantify improvements in accuracy, response time, and usability.
- **Practical Impact:** Integration with Power BI and other enterprise tools shows real-world applicability, bridging the gap between theoretical research and industrial deployment.

Overall, while converting this work into a full research paper is not mandatory, the project contains sufficient novelty, technical depth, and evaluative potential to support publication in AI, HCI, or business analytics venues.

17 **Conclusion**

IntelliQuery represents a significant advancement in business intelligence accessibility, combining cutting-edge conversational AI with practical enterprise needs. The project addresses critical barriers that prevent non-technical users from effectively leveraging business data, while contributing valuable research insights to the fields of conversational AI and human-computer interaction.

Through our industry partnership with Folio3, the project ensures real-world applicability and enterprise-grade quality. The comprehensive technical approach, involving advanced algorithms, custom dashboard development, and sophisticated AI integration, demonstrates the practical application of computer science principles to solve meaningful business problems.

The expected impact includes democratized access to business intelligence, significant time savings in data analysis tasks, and a reusable framework for conversational BI applications. With its combination of technical innovation, academic rigor, and industry relevance, IntelliQuery positions itself as both a valuable commercial solution and a significant contribution to the research community.

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